OOP Assignment: 01

Topics: Encapsulation, this, super, Abstract Class and Abstract Method, Interface

Assignment 01:

Write a program using any object-oriented language (e.g., Java, C++, or Python) to represent a Bank Account.

Your program should include:

- A class BankAccount with data members for account holder's name, account number, and balance.
- Member functions deposit(), withdraw(), and display() to perform banking operations.
- Ensure that the withdrawal operation does not allow withdrawing more money than the available balance.
- Demonstrate encapsulation by keeping the data members private and providing appropriate public methods to access and modify them.

Assignment 02:

Write a program in Java to represent an employee using a class named Employee.

Your program should include:

- Data members employeeId, employeeName, and salary.
- A parameterized constructor to initialize these data members.
- Use the **this** keyword to distinguish between instance variables and constructor parameters having the same names.
- A method displayDetails() to display the employee's information.
- Create at least two employee objects and display their details.

Assignment 03:

Write a program in Java to represent a student using a class named Student.

Your program should include:

- Private data members studentId, studentName, and marks.

- A parameterized constructor that initializes the data members using the this keyword.
- Public getter and setter methods to access and modify the private data members.
- Ensure that the marks cannot be less than 0 or greater than 100.
- A method displayResult() to display the student's information and marks.

Assignment 04:

Write a program in Java to demonstrate inheritance and the use of the super keyword.

Your program should include:

- A base class Vehicle with data members brand and speed.
- A parameterized constructor to initialize these data members.
- A derived class Car that inherits from Vehicle and has an additional data member numberOfDoors.
- Use the super keyword to call the constructor of the base class.
- Create an object of the Car class and display all vehicle information.

Assignment 05:

Write a program in Java to demonstrate the use of the super keyword in inheritance.

Your program should include:

- A base class Person with data members name and age.
- A method display() to display the person's information.
- A derived class Student that inherits from Person and contains an additional data member studentId.
- Override the display() method in the Student class.
- Use super.display() to call the parent class's method from the overridden method.
- Create a Student object and display all information.

Assignment 06:

Write a program in Java to demonstrate abstraction using an abstract class and an abstract method.

Your program should include:

- An abstract class Shape with an abstract method calculateArea().
- Two derived classes, Rectangle and Circle, that inherit from Shape.
- Implement the calculateArea() method in the Rectangle class to calculate the area using length and width.
- Implement the calculateArea() method in the Circle class to calculate the area using the formula (pi * r^2).
- Create objects of both derived classes and display their respective areas.

Assignment 07:

Write a program in Java to demonstrate abstraction using an abstract class and abstract methods.

Your program should include:

- An abstract class Employee with data members employeeId and employeeName.
- Declare an abstract method calculateSalary().
- Create two derived classes, PermanentEmployee and ContractEmployee, that inherit from Employee.
- Implement calculateSalary() in both classes using the following rules:
    - A permanent employee receives a fixed monthly salary of Tk. 40,000.
    - A contract employee receives Tk. 1,500 per working day.
- Create objects of both classes and display their details and calculated salaries.

Assignment 08:

Your program should include:

- Create an interface named SalaryCalculator with a method calculateSalary().
- Create two classes, PermanentEmployee and ContractEmployee, that implement the SalaryCalculator interface.

- The PermanentEmployee class should calculate a fixed monthly salary of Tk. 40,000.
- The ContractEmployee class should calculate salary based on the number of working days at Tk. 1,500 per day.
- Implement the calculateSalary() method in both classes.
- Create objects of both classes in the main() method and display their calculated salaries.